Jay Collins

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Education

University of Colorado Boulder, Boulder, CO
B.S. Electrical Engineering Expected May 2027
GPA 3.82, Dean's List

Technical Courses

- Circuits I & II (A), Linear Systems (A), Electronics Design Lab (A–), Embedded Software Engineering (A)
- Data Structures (A), Programmable Digital Systems (A), Microelectronics (A), Fields and Waves (A–)
- Calculus I (A–), Calculus II (A), Calculus III (A–), Probability and Statistics (A)

Skills

Languages: C, C++, Python, RISC-V Assembly, Verilog, LaTeX

Protocols & Interfaces: SPI, I2C, UART, I2S, PWM, GPIO Interrupts, Bluetooth, WiFi, REST APIs

Software & Debugging: FreeRTOS (RTOS), STM32CubeIDE, GDB, UART/Serial Logging, Altium, LTspice, VS Code, GitHub

Hardware & Testing: PCB Design, SMD, THT, Soldering, Oscilloscopes, Function Generators, DMM, Power Supplies

Core Engineering Skills: Circuit Design, Embedded Systems, Digital Logic, State Machines, IoT

Professional Skills: Technical Communication, Team Collaboration, Documentation

Experience

Engineering Learning Assistant – CU Boulder Jan 2026 – Present

- Learning Assistant for Linear Systems and Exploring ECE, supporting 150+ students in weekly office hours.
- Explained complex concepts like convolution and Fourier transforms in clear, accessible language.

Deckhand – Hawai'i Nautical May 2024 – Present

- Crew member aboard a 54 ft sailing and snorkeling catamaran, handling bartending, general maintenance, and passenger safety during tours.
- Assisted with maintenance and monitoring of onboard electrical systems, applying hands-on problem-solving to identify and address issues before they impacted operations. Lifeguard and CPR certified.

Projects

documented at github.com/2jaycollins

World's Worst Alarm Clock – WiFi-Connected Embedded System

- Built an ESP32 alarm clock in C++ with a real-time state machine, interrupt-driven KY-040 encoder/button inputs, SPI SSD1309 OLED refreshing at 30 Hz, I2C DS3231 RTC with NTP sync, and live weather via REST API.
- Moved all networking into a dedicated FreeRTOS task on the ESP32's second core, using task notifications and a mutex to share the radio with Bluetooth audio, so network delays never stall the UI or the alarm.
- Added Bluetooth audio through a MAX98357 I2S amplifier, and took the hardware from breadboard to soldered protoboard to a custom 3D-printed enclosure.

BLDC Controlled Driving Robot

- Built a BLDC driving robot on an Arduino Nano in C++, playing music with `tone()` PWM while executing driving patterns with under 1 cm of positional error.
- Prototyped the motor control and audio circuits on a breadboard, running both from a single microcontroller.

STM32 Casino Game

- Built blackjack and roulette in C on an STM32F429 Discovery board, with interrupt-driven touch input responding in under 20 ms.
- Generated game outcomes with the STM32 hardware RNG via the HAL; debugged firmware with GDB in STM32CubeIDE.